

3.3 Halogen Compounds

Question Paper

Course	CIE A Level Chemistry
Section	3. Organic Chemistry
Topic	3.3 Halogen Compounds
Difficulty	Easy

Time allowed: 30

Score: /24

Percentage: /100

Question 1a

The structure of a halogenoalkane containing bromine is shown in Fig. 1.1.

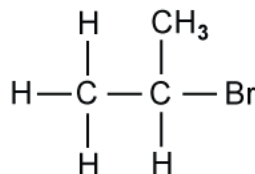


Fig. 1.1

i)

Name the compound shown in Fig. 1.1.

[1]

ii)

State the class of halogenoalkane to which this compound belongs.

[1]

[2 marks]

Question 1b

State the colour of the precipitate formed in the reaction between the halogenoalkane in Fig. 1.1 in part (a) and acidified silver nitrate, AgNO_3 .

[1 mark]

Question 1c

The compound drawn in Fig. 1.1 in part (a) is reacted with alcoholic potassium cyanide, KCN. The reaction is heated under reflux.

Draw the product of this nucleophilic substitution reaction.

[1 mark]

Question 1d

The halogenoalkane drawn in Fig. 1.2 contains a chlorine atom instead of a bromine atom. This halogenoalkane is reacted under the same conditions outlined in part (c).

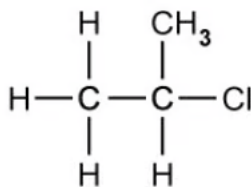


Fig. 1.2

The reaction for a halogenoalkane containing bromine would be faster than if the compound in Fig. 1.2 was used. Briefly explain why this reaction would be faster.

[2 marks]

Question 2a

Fig. 2.1 shows the reaction profile for the production of propene.



Fig. 2.1

- i)
State the conditions required for step 1.

[2]

- ii)
Name the mechanism for step 1.

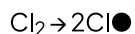
[1]

[3 marks]

Question 2b

The mechanism for step 1 in part (a) involves three different steps, initiation, propagation and termination.

Initiation:

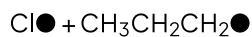


Propagation:

Equation 1

Equation 2

Termination:



The two propagation steps are missing. Write both of the equations that are required for this step.

[2 marks]

Question 2c

i)

Name the mechanism for step 2.

[1]

ii)

State the reagent and necessary conditions for step 2.

[3]

[4 marks]

Question 3a

Many of the reactions of halogenoalkanes involve a nucleophile attacking the carbon attached to the halogen atom. The nucleophile replaces the halogen atom in a nucleophilic substitution reaction.

The mechanism for the reaction is determined by the structure of the halogenoalkane. Primary and tertiary halogenoalkanes react via different reaction mechanisms.

i)

State what is meant by the term tertiary halogenoalkane.

[1]

ii)

Tertiary halogenoalkanes react via an S_N1 mechanism, whereas primary halogenoalkanes react via an S_N2 mechanism.

Explain what the numbers 1 and 2 refer to in S_N1 and S_N2 .

[1]

[2 marks]

Question 3b

2-bromo-2-methylpropane reacts with hydroxide ions via an S_N1 mechanism.

i)

The incomplete reaction mechanism for the reaction is shown in Fig. 3.1

Complete the reaction mechanism shown in Fig. 3.1 clearly showing any partial charges, full charges and the transfer of electrons.

[3]

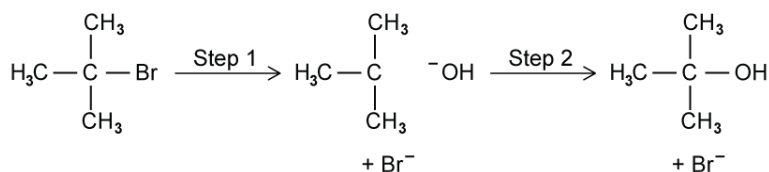


Fig. 3.1

ii)

Identify which, step 1 or step 2, is the rate-determining step.

[1]

[4 marks]

Question 3c

An ethanolic solution of excess ammonia (NH_3 in ethanol) is heated under pressure with 2-bromo-2-methylpropane.

i)

Draw the structure of the resulting organic product.

[1]

ii)

State the name of the functional group of the organic product.

[2]

[3 marks]

Model Answers: Easy

1a

a)

i) The name of the halogenoalkane is:

- 2-bromopropane; [1 mark]

ii) The class of halogenoalkane is:

- Secondary; [1 mark]

[Total: 2 marks]

- The Br atom is bonded to a C atom which is bonded to two alkyl groups
- This makes the compound a secondary halogenoalkane

1b

b) The colour of the precipitate formed is:

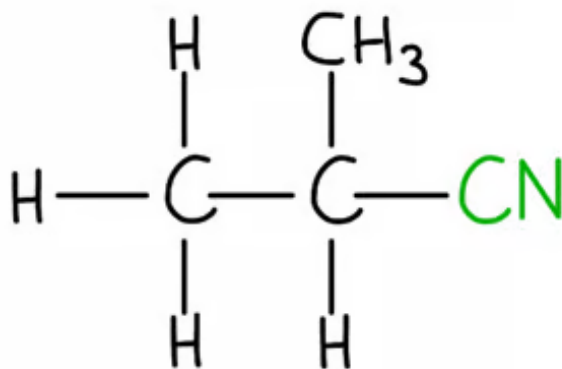
- Cream (precipitate); [1 mark]

[Total: 1 mark]

- **Remember:** Dilute ammonia and concentrated ammonia are used to further distinguish between the silver halide precipitates
 - **White AgCl (s)** is soluble in dilute ammonia
 - **Cream AgBr (s)** is insoluble in dilute ammonia but soluble in concentrated ammonia
 - **Yellow AgI (s)** is insoluble in concentrated ammonia

1c

c) The product of the reaction is:



- Correctly drawn product; [1 mark]

[Total: 1 mark]

- This reaction will involve nucleophilic substitution
- The Br atom will be substituted by the CN^- nucleophile

1d

d) The reaction with the bromine containing halogenoalkane would be faster because:

- The C-Cl bond is stronger (than the C-Br bond); [1 mark]
- (Therefore) takes more energy to break; [1 mark]

[Total: 2 marks]

- The strength of the carbon-halogen bonds decreases as you go down Group 17
 - $\text{C-Cl} > \text{C-Br} > \text{C-I}$
- Therefore the rate of the substitution reaction will increase as you go down Group 17
 - $\text{C-I} > \text{C-Br} > \text{C-Cl}$
- One way of measuring the rate of this reaction is to use a halide test:
 - The halide ion is substituted and will form a precipitate with silver nitrate solution
 - The quicker the precipitate forms, the faster the rate of reaction

2a

a)

i) The conditions required for step 1 are:

- UV light; [1 mark]
- Chlorine / Cl_2 ; [1 mark]

ii) The mechanism for step 1 is:

- Free radical substitution; [1 mark]

[Total: 3 marks]

- To substitute an H atom for a Cl atom in an alkane, UV light is required
- The Cl_2 molecule is split into two Cl free radicals

2b

b) Both propagation steps are:

- $\text{Cl}\bullet + \text{CH}_3\text{CH}_2\text{CH}_3 \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\bullet + \text{HCl}$; [1 mark]
- $\text{CH}_3\text{CH}_2\text{CH}_2\bullet + \text{Cl}_2 \rightarrow \text{CH}_3\text{CH}_2\text{CH}_2\text{Cl} + \text{Cl}\bullet$; [1 mark]

[Total: 2 marks]

- It is important to remember that there are two steps for propagation
- **Note:**
 - There is no net change in the number of free radicals
 - The chlorine free radical used in the first propagation step is regenerated in the second step

2c

c)

i) The name of the mechanism for step 2 is:

- Elimination; [1 mark]

ii) The reagent and necessary conditions for step 2 are:

- Ethanol; [1 mark]
- (Concentrated) sodium hydroxide / NaOH ; [1 mark]
- Heat; [1 mark]

[Total: 4 marks]

- Elimination involves the removal of an H and Cl atom to produce an alkene with the elimination of the small molecule HCl
- You need to know the necessary conditions for this reaction
- Potassium hydroxide can be used instead of sodium hydroxide and it would gain a mark
- The use of ethanolic sodium hydroxide is very important as if aqueous sodium hydroxide is used, a nucleophilic substitution reaction occurs and an alcohol would be produced instead of an alkene:
 - 1-chloropropane + sodium hydroxide (aq) $\rightarrow$ propan-1-ol + sodium chloride

3a

a)

i) The term tertiary halogenoalkane means:

- The carbon atom bonded to the halogen atom is bonded to 3 other carbon atoms / alkyl groups; **[1 mark]**

ii) The numbers 1 and 2 in S_N1 and S_N2 refer to:

- The number of reacting species in the slow step / rate-determining step
OR
2 shows that the slow step / rate determining step involves two reacting species
AND
1 shows that the slow step / rate determining step involves one reacting species; **[1 mark]**

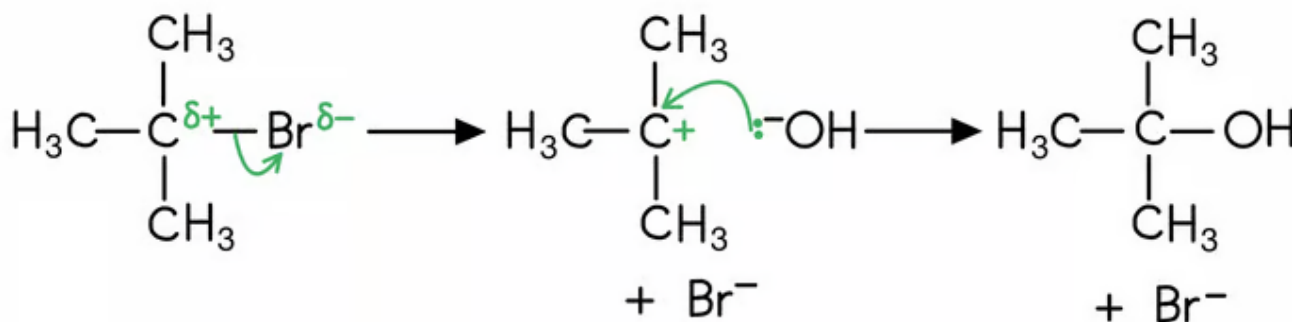
[Total: 2 marks]

- You should be able to identify primary, secondary and tertiary halogenoalkanes from their structures
- You should also be aware of which type of mechanism is applicable to primary and tertiary halogenoalkanes
- **Remember:** Secondary halogenoalkanes will react by a mixture of the two mechanisms, depending on their structure, but you are not expected to know any further details about the reaction mechanism of secondary halogenoalkanes

3b

b)

i) The completed reaction mechanism is:



- δ^+ on C attached to Br and δ^- on Br **AND** curly arrow from C-Br bond to Br; [1 mark]
- Intermediate with + charge; [1 mark]
- Lone pair on OH^- **AND** curly arrow from lone pair to correct C atom; [1 mark]

ii) The rate-determining step is:

- Step 1; [1 mark]

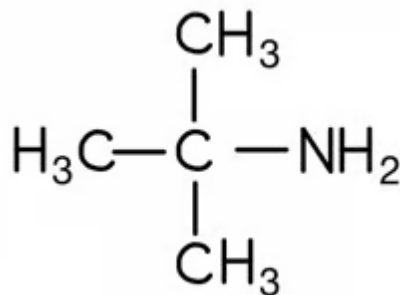
[Total: 4 marks]

- In the $\text{S}_{\text{N}}1$ mechanism, the first step is to break the C-Br bond
 - This is the rate-determining step as it is the slowest step; it only involves the halogenoalkane i.e. one reacting species
- A tertiary carbocation is formed
- The OH^- ion acts as a nucleophile as it donates a pair of electrons to the carbocation
- You should be able to draw this mechanism - remember to include any dipoles, lone pairs and any curly arrows should either start from a lone pair of electrons or a bond

3c

c)

i) The structure of the resulting organic product is:



- Correct structure; [1 mark]

ii) The name of the functional group of the organic product is:

- Amine (group); [1 mark]

[Total: 2 marks]

- When an ethanolic solution of excess ammonia (NH_3 in ethanol) is heated under pressure with the halogenoalkane, an amine is formed:
 - $\text{CH}_3\text{C}(\text{CH}_3)_2\text{Br} + \text{NH}_3 \rightarrow \text{CH}_3\text{C}(\text{CH}_3)_2\text{NH}_2 + \text{HBr}$
- Amines have the $-\text{NH}_2$ functional group
- This is another example of a nucleophilic substitution reaction where ammonia acts as the nucleophile